Lecture 1, Dec 17, 2025: Inequalities and Absolute Value.

In the real numbers we have two possibilities for two arbitrary numbers.

i). The numbers are equal "$=$"

for example: $1, \underbrace{\frac{5}{5}}_{=}$ or $2, \underbrace{\sqrt[3]{8}}_{=}$

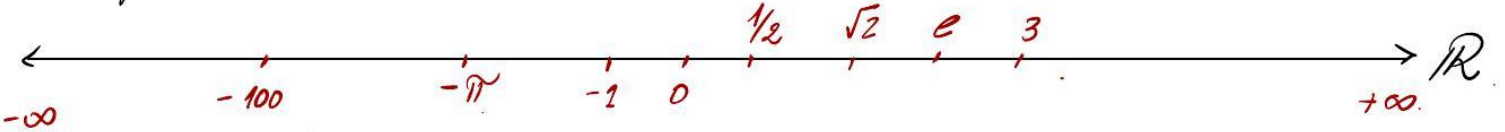
ii). The numbers are different, but in this case we can ask yourself about which is greater than the another or less than the other.

In general if x, y are real numbers then:

$$x = y \quad \text{or} \quad x < y \quad \text{or} \quad x > y.$$

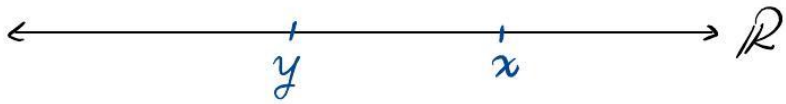
"x equal to y" "x less than y" "x greater than y"

Using this property we can draw all the real numbers in a number line representation.

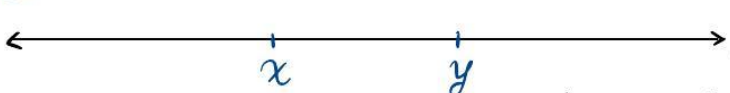


In this line there are all the real numbers and we can give an interpretation for $x > y$ or $x < y$ geometrically.

We say $x > y$ if x is in the right side of y

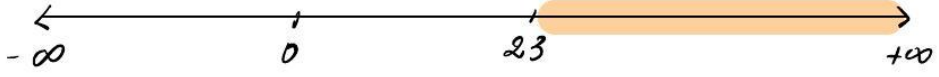


We say $x < y$ if x is in the left side of y.



The symbols $x \geq y$ and $z \leq y$ are true if $x = y$ or $x > y$ and $z = y$ or $z < y$

If you fix 23, then which are the numbers greater than 23?



We have infinite numbers.

Def (Set). A set is a collection (sometimes infinite) of objects, and the objects are call the elements.

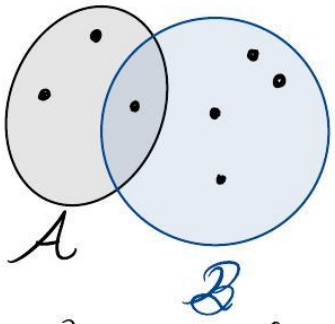
Notation: • If S is a set the notation $a \in S$ means a is an elements of S, otherwise $a \notin S$.

For example: $\mathbb{N}$ = natural numbers = $\{0, 1, 2, 3, 4, 5, \cdots\}$.

Notice that $5 \in \mathbb{N}$ but $\sqrt{2} \notin \mathbb{N}$.

Operations Between Sets

i) **Union:** If A and B are two sets then the union $A \cup B$ is the set consisting of elements that are in A or B.



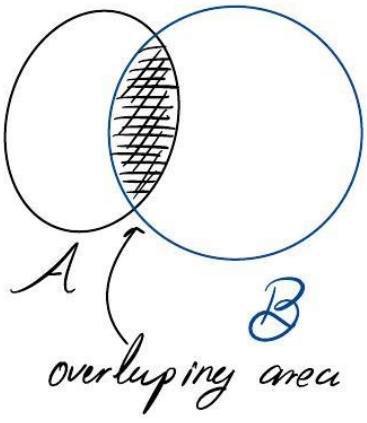
$A \cup B$ = everything.

Example: $A = \{1, 2, \square\}$ $B = \{\star, \frac{1}{2}, -1\}$.

then

$$A \cup B = \{1, 2, \square, \star, \tfrac{1}{2}, -1\}$$

ii) **Intersection:** If A and B are two sets then $A \cap B$ is the set consisting of element in A and B



Example: $A = \{\bigstar, 1, \pi, e\}$ $B = \{7, 32, -15, \bigstar\}$.

$$A \cap B = \{\bigstar\}$$

If A and B don't have common elements then $A \cap B = \emptyset$.

$\emptyset$ is the set that contains no elements

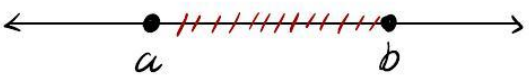
More examples of sets:

$A = \{1, 2, 3, 4, 5, 6\}$ in other notation

$A = \{x \mid x \text{ is natural number and } 0 < x < 7\}$

A particular case of sets:

Intervals:

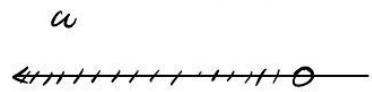
	Geometrically	Set Notation
Open interval:		$(a,b) = \{x \in \mathbb{R} \mid a < x < b\}$
Closed interval		$[a,b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$

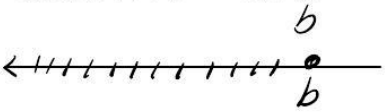
Combinations of these intervals

$(a,b] =$  $= \{x \in \mathbb{R} \mid a < x \leq b\}$.

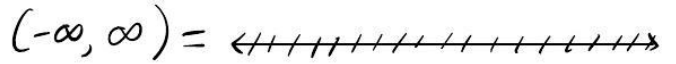
$[a,b) =$  $= \{x \in \mathbb{R} \mid a \leq x < b\}$.

$(a,\infty) =$  $= \{x \in \mathbb{R} \mid x > a\}$.

$[a,\infty) =$  $= \{x \in \mathbb{R} \mid x \geq a\}$.

$(-\infty, b) =$  $= \{x \in \mathbb{R} \mid x < b\}$

$(-\infty, b] =$ $= \{x \in \mathbb{R} \mid x \leq b\}$.

$(-\infty, \infty) =$  $= \mathbb{R}$.

Problem: A bus can carry a maximum of 65 passengers or less. If on tip, the bus already carries 30 people, how many more people can get on the bus?

Model: Let x the number of additional people. So,

$$30 + x \leq 65 \text{ (an inequality)}$$

Goal: We want describe x.

Rules for inequalities:

i) $a < b$ then $a + c < b + c$.

ii) if $a < b$ and $c < d$ then $a + c < b + d$.

iii) if $a < b$ and $c > 0$ then $ac < bc$.

iv) if $a < b$ and $c < 0$ then $ac > bc$

v) if $0 < a < b$ then $\frac{1}{b} < \frac{1}{a}$

with these properties we can solve inequalities.

$$x + 30 \leq 65$$

From i) $c = -30$ then

$$x + 30 + (-30) \leq 65 + (-30)$$

$$x \leq 65 - 30$$

$$x \leq 35$$

So, **Ans:** 35 or fewer people can be on the bus.

Example: Solve $1 + x < 7x + 5$

Sln: **Step 1:** All the variables in one side (left or right).

Hint: try keep the coefficient of the variable positive.

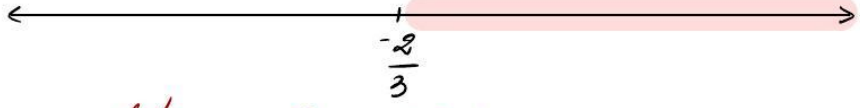
$$1 < 7x + 5 - x$$

$$1 - 5 < 7x - x$$

$$-4 < 6x$$

$$\frac{-4}{6} < x.$$

$$-\frac{2}{3} < x. \qquad \text{So}$$



$$S = \left(-\frac{2}{3}, \infty\right).$$

Solve the inequality: $x^2 - 5x + 6 \leq 0$.

Step 1 Unequal to zero (this step is free here)

Step 2 factorize

$$x^2 - 5x + 6 \leq 0$$

$$(x-3)(x-2) \leq 0.$$

Step 3. Draw a line with the roots of the polynomials above.


+ + + +

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Problem. Solve

$$\underbrace{|3x-2|}_{a} = \begin{cases} \underbrace{3x-2}_{a} & \text{if } \underbrace{3x-2}_{a} \geq 0 \\ \underbrace{-(3x-2)}_{-a} & \text{if } \underbrace{3x-2}_{a} < 0 \end{cases}$$

Solving $3x-2 \geq 0$ and $3x-2 < 0$

$3x \geq 2$ and $3x < 2$

$x \geq \frac{2}{3}$ and $x < \frac{2}{3}$

$$|3x-2| = \begin{cases} 3x-2 & \text{if } x \geq \frac{2}{3} \\ 2-3x & \text{if } x < \frac{2}{3} \end{cases}$$

Properties: i). $|a \cdot b| = |a| \cdot |b|$

ii). $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ if $b \neq 0$.

iii). $|a^n| = |a|^n$

Now assume $a > 0$ then.

i). $|x| = a$ if and only if $x = \pm a$.

ii). $|x| < a$ if and only if $-a < x < a$. ($-a < x$ and $x < a$)

iii). $|x| > a$ if and only if $x > a$ or $x < -a$

Example: i). $|2x-5| = 3 \overset{\text{def}}{\iff} 2x-5 = 3$ or $2x-5 = -3$

$2x = 3+5$ $\quad$ $2x = -3+5$

$2x = 8$ $\quad$ $2x = 2$

$x = \frac{8}{2}$ $\quad$ $x = \frac{2}{2}$

$x = 4$ $\quad$ $x = 1$

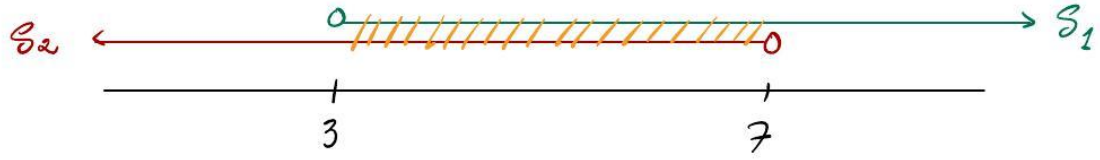
ii). $|x-5|<2$ $\overset{\text{def}}{\Longleftrightarrow}$ $-2<x-5$ and $x-5<2$.

$-2+5<x$ $\qquad x<2+5$

$3<x$ $\qquad x<7$

$S_1=(3,\infty)$ intersection $S_2=(-\infty,7)$.

$S=S_1\cap S_2$



$S=(3,7)$.